THE IMAGINARY PARTS OF THE CONVERGENTS OF $[1+i;\,1+i,\,1+i,\ldots]$ ARE $\mathrm{A317974}(n+1)/\mathrm{A138573}(n)$

MATH0

Abstract. Let c_n be the n-th convergent of the continued fraction $(1+i)+1/((1+i)+1/((1+i)+\cdots))$ and write $c_n = (\mathrm{A093725}(n)+\mathrm{A093726}(n)\,i)/\mathrm{A093727}(n)$, where $\mathrm{A093727}(n)$ is the least positive integer d with $dc_n \in \mathbb{Z}[i]$. We prove Zhuorui He's 2026 conjecture from OEIS A093726 that $\mathrm{A093726}(n)/\mathrm{A093727}(n) = \mathrm{A317974}(n+1)/\mathrm{A138573}(n)$ for all $n \geq 1$, i.e. $\operatorname{Im} c_n = \mathrm{A317974}(n+1)/\mathrm{A138573}(n)$, where both sequences satisfy $u(n) = 2(u(n-1)+u(n-2)+u(n-3))-u(n-4)$. The proof writes $c_n = U(n+1)/U(n)$ for the Gaussian continuants U and identifies $|U(n)|^2$ (as observed by P. Bala for A138573) and $\operatorname{Im}\big(U(n+1)\overline{U(n)}\big)$ with the two recurrence sequences. We know of no earlier proof, so we classify the result as new, although the argument is elementary. The theorem is verified in Lean 4 against the frozen, fidelity-reviewed math0 formalization of the conjecture, whose definitions are checked by the kernel against the OEIS data.

1. Introduction

OEIS A093726 [7], contributed by Robert G. Wilson v on March 11, 2004, lists the numerators of the imaginary parts of the convergents of the periodic Gaussian continued fraction

$$(1+i)+\cfrac{1}{(1+i)+\cfrac{1}{(1+i)+\cdots}};$$

its companions A093725 and A093727 [6] list the numerators of the real parts and the common denominators. The convergents begin $1+i$, $(3+i)/2$, $(8+4i)/5$, $(6+3i)/4$, $(23+11i)/15$. On June 16, 2026, Zhuorui He corrected the definition of A093726, extended its data to 30 terms, added these worked examples, and added the conjecture

$$\mathrm{A093726}(n)/\mathrm{A093727}(n) = \mathrm{A317974}(n+1)/\mathrm{A138573}(n) \qquad (n \geq 1),$$

in which A317974 [9] (N. J. A. Sloane, 2018) and A138573 [8] (B. Cloitre, 2008) are the solutions of the recurrence with signature $(2,2,2,-1)$,

$$u(n) = 2u(n-1)+2u(n-2)+2u(n-3)-u(n-4) \qquad (n \geq 4), \tag{1}$$

with initial terms $0,0,1,1$ and $0,1,2,5$ respectively. We prove the conjecture (Theorem 2).

Prior work and status. In A138573, Peter Bala (March 4, 2014) noted that $\mathrm{A138573}(n) = |U(n)|^2$ for the Lucas sequence $U(0)=0$, $U(1)=1$, $U(n) = (1+i)U(n-1)+U(n-2)$ in the Gaussian integers [8]. Since $U(n)$ is the n-th continuant of the continued fraction, this explains the denominators $\mathrm{A138573}(n)$ in the conjecture. We found no statement, in OEIS or elsewhere, connecting A317974 with these convergents; besides its recurrence and programs, the entry A317974 contains a generating function (C. Barker), a growth rate (A. H. M. Smeets) and links to Coxeter's paper on loxodromic sequences of tangent spheres [1] and to MathWorld. As of revision #13 of A093726 (June 20, 2026) the formula is still marked as a conjecture, and no later edit records a proof. The math0 referees searched arXiv, MathOverflow, Mathematics Stack

Date: October 24, 2026.

Exchange and the SeqFan archives and found no proof or counterexample; the recent collection of proofs of OEIS conjectures by S. Fried [3] does not treat A093726, A093727, A317974 or A138573. We therefore classify the result as new. It is elementary: the new ingredient, beyond the classical continuant formula $c_n = U(n+1)/U(n)$ and Bala's observation, is the identification of $\operatorname{Im}\big(U(n+1)\overline{U(n)}\big)$ with A317974$(n+1)$.

Method. Put $S(n) = |U(n)|^2$ and $P(n) + iQ(n) = U(n+1)\overline{U(n)}$. Then $c_n = (P(n) + iQ(n))/S(n)$ for $n \geq 1$. Three integer identities relating P, Q, S at n and $n+1$ drive an induction for this closed form, and imply that S and Q satisfy (1); comparing four initial values gives $S(n) = \text{A138573}(n)$ and $Q(n) = \text{A317974}(n+1)$. Finally, the left-hand side of the conjecture is exactly $\operatorname{Im} c_n$.

2. Statement

For $x \in \mathbb{Q}$ let $\operatorname{num} x \in \mathbb{Z}$ and $\operatorname{den} x \geq 1$ be its numerator and denominator in lowest terms. For a Gaussian rational $z = x + yi$ $(x, y \in \mathbb{Q})$ let

$$\operatorname{den} z = \operatorname{lcm}(\operatorname{den} x, \operatorname{den} y), \qquad \operatorname{num} z = (\operatorname{den} z)\, z \in \mathbb{Z}[i].$$

Thus $\operatorname{den} z$ is the least positive integer d with $dz \in \mathbb{Z}[i]$; this is the convention of Mathematica's `Numerator` and `Denominator` for Gaussian rationals.

Definition 1. Let $c_1 = 1+i$ and $c_{n+1} = (1+i)+1/c_n$ for $n \geq 1$, so that $c_n = [1+i; 1+i, \ldots, 1+i]$ has n partial quotients. (This is well defined: if $\operatorname{Re} c_n \geq 1$ then $\operatorname{Re}(1/c_n) = \operatorname{Re} c_n/|c_n|^2 > 0$, so $\operatorname{Re} c_{n+1} \geq 1$.) Set

$$\text{A093725}(n) = \operatorname{Re} \operatorname{num} c_n, \qquad a(n) = \text{A093726}(n) = \operatorname{Im} \operatorname{num} c_n, \qquad \text{A093727}(n) = \operatorname{den} c_n.$$

Let A317974 and A138573 be the sequences (offset 0) satisfying (1) with initial terms $0, 0, 1, 1$ and $0, 1, 2, 5$.

For example, $c_4 = \frac{3}{2} + \frac{3}{4}i = (6+3i)/4$, so A093725$(4) = 6$, $a(4) = 3$, A093727$(4) = 4$, as in the examples of A093726. The FORMULA section of A093726 reads, verbatim:

> `Conjecture: a(n)/A093727(n) = A317974(n+1)/A138573(n). - _Zhuorui He_, Jun 16 2026`

Theorem 2. *For every $n \geq 1$,*

$$\frac{a(n)}{\text{A093727}(n)} = \frac{\text{A317974}(n+1)}{\text{A138573}(n)}$$

as rational numbers.

In the proof (Section 3) we show that both sides equal $\operatorname{Im} c_n$.

Lean formalization. The statement was formalized by math0 in `lean/Math0/Statements/OEIS/A093726.lean`, reviewed for fidelity, and frozen (SHA-256 `5d599ac4...08f83a0ba`) before any proof attempt. Its definitions and the statement read as follows (verbatim; docstrings and the data theorems are omitted):

```
namespace Math0.Statements.A093726
def gadd (z w : ℚ × ℚ) : ℚ × ℚ := (z.1 + w.1, z.2 + w.2)
def ginv (z : ℚ × ℚ) : ℚ × ℚ :=
  (z.1 / (z.1 * z.1 + z.2 * z.2), -z.2 / (z.1 * z.1 + z.2 * z.2))
def cfValue : List (ℚ × ℚ) → ℚ × ℚ
  | [] => (0, 0)
  | [x] => x
  | x :: xs => gadd x (ginv (cfValue xs))
```

```
def conv (n : ℕ) : ℚ × ℚ := cfValue (List.replicate n (1, 1))
def gden (z : ℚ × ℚ) : ℕ := Nat.lcm z.1.den z.2.den
def gnumIm (z : ℚ × ℚ) : ℤ := z.2.num * ((gden z / z.2.den : ℕ) : ℤ)
def a (n : ℕ) : ℤ := gnumIm (conv n)
def A093727 (n : ℕ) : ℕ := gden (conv n)
def A317974 : ℕ → ℤ
  | 0 => 0
  | 1 => 0
  | 2 => 1
  | 3 => 1
  | n + 4 => 2 * (A317974 (n + 3) + A317974 (n + 2) + A317974 (n + 1)) - A317974 n
def A138573 : ℕ → ℤ
  | 0 => 0
  | 1 => 1
  | 2 => 2
  | 3 => 5
  | n + 4 => 2 * A138573 (n + 3) + 2 * A138573 (n + 2) + 2 * A138573 (n + 1) - A138573 n
def statement : Prop :=
  ∀ n : ℕ, 1 ≤ n →
    (a n : ℚ) / (A093727 n : ℚ) = (A317974 (n + 1) : ℚ) / (A138573 n : ℚ)
```

How the definitions encode the entry. A Gaussian rational $x+yi$ is the pair $(x, y) \in \mathbb{Q}\times\mathbb{Q}$; `gadd` is addition and `ginv` is $1/(x+yi) = (x-yi)/(x^2+y^2)$ (Lean's division by 0 returns 0, but `ginv` is only ever applied to nonzero values here). `cfValue` evaluates $x_1 + 1/(x_2 + \cdots + 1/x_n)$, like Mathematica's `FromContinuedFraction`, and `conv n` follows the entry's program `Table[Im[Numerator[FromContinuedFraction[Table[1+I,{n}]]]],{n,30}]`. Since `List.replicate (n+2) (1, 1)` is $(1,1)$ followed by a nonempty list, `conv (n + 2)` unfolds to `gadd (1, 1) (ginv (conv (n + 1)))`, and `conv 1 = (1, 1)`; so `conv n` is c_n for $n \geq 1$. `gden` is $\operatorname{den} z$, and since $\operatorname{den} y \mid \operatorname{den} z$ the natural-number quotient in `gnumIm` is exact, so `gnumIm z` $= \operatorname{num} y \cdot (\operatorname{den} z/\operatorname{den} y) = \operatorname{Im}\operatorname{num} z$. The sequences A317974 and A138573 are their defining recurrences, with offset 0; A093726 and A093727 have offset 1, matching the hypothesis $1 \leq n$. The statement is an equality in $\mathbb{Q}$, and both denominators are positive.

The file checks in the kernel (`decide +kernel`) that `a` reproduces all 30 listed terms of A093726, `A093727` all 28 listed terms of A093727, and `A317974`, `A138573` their first 20 listed terms (theorems `a_data`, `A093727_data`, `A317974_data`, `A138573_data`). No instance of the conjecture is checked in the statement file.

Remark 3 (Reading of the name)**.** The name of A093726 ("the numerator of the imaginary part of the convergents") could also be read as $\operatorname{num}(\operatorname{Im} c_n)$. The formalization uses $\operatorname{Im}\operatorname{num} c_n$, which is what the entry's program computes and what its examples use ("(6+3*i)/4, so A093725(4)=6, a(4)=3 and A093727(4)=4"). The two readings agree on all listed terms but first differ at $n = 42$ (computer check, not formalized): $\operatorname{Im} c_{42} = 18854485653829/25378408048970$ in lowest terms, while $A093727(42) = 329919304636610 = 13 \cdot 25378408048970$, so $a(42) = 13 \cdot 18854485653829$. Under the other reading the left side of the conjecture would be $\operatorname{Im} c_{42}/13$, and the conjecture would fail at $n = 42$.

3. Proof

We follow the Lean proof in `lean/Math0/Frontier/OEIS/A093726.lean`; the names of its (private) lemmas are given in brackets.

3.1. Continuants.

Define Gaussian integers $U(n)$, stored in Lean as integer pairs, by

$$U(0) = 0, \qquad U(1) = 1, \qquad U(k+2) = (1+i)\,U(k+1) + U(k)$$

(`U`, `step`), and put

$$S(n) = |U(n)|^2, \qquad P(n) + iQ(n) = W(n) := U(n+1)\,\overline{U(n)} \qquad (P(n), Q(n) \in \mathbb{Z})$$

(`S`, `P`, `Q`). In coordinates, if $U(n) = x_0 + y_0 i$ and $U(n+1) = x_1 + y_1 i$, then $P(n) = x_1 x_0 + y_1 y_0$ and $Q(n) = y_1 x_0 - x_1 y_0$. The first values are

n	0	1	2	3	4	5	6
$U(n)$	0	1	$1+i$	$1+2i$	$4i$	$-3+6i$	$-9+7i$
$S(n)$	0	1	2	5	16	45	130
$P(n)$	0	1	3	8	24	69	199
$Q(n)$	0	1	1	4	12	33	97

Lemma 4. *For every $n \geq 0$:*

(a) $P(n+1) = S(n+1) + P(n)$;
(b) $Q(n+1) = S(n+1) - Q(n)$;
(c) $P(n)^2 + Q(n)^2 = S(n)\,S(n+1)$;
(d) $S(n+2) = 2S(n+1) + S(n) + 2P(n) - 2Q(n)$.

Proof. We have $W(n+1) = \big((1+i)U(n+1) + U(n)\big)\overline{U(n+1)} = (1+i)S(n+1) + \overline{W(n)}$; its real and imaginary parts give (a) and (b). (c) is $|W(n)|^2 = |U(n+1)|^2 |U(n)|^2$. For (d), expand, using $|1+i|^2 = 2$, $S(n+2) = |(1+i)U(n+1) + U(n)|^2 = 2S(n+1) + S(n) + 2\,\mathrm{Re}\big((1+i)W(n)\big)$ and use $\mathrm{Re}((1+i)(P+iQ)) = P - Q$. In Lean, (a) and (b) are proved inside `conv_eq` by unfolding $U(n+2)$ through `step` and normalizing with `ring`, and (c) by `ring` (the Lagrange identity $(x_1x_0 + y_1y_0)^2 + (y_1x_0 - x_1y_0)^2 = (x_0^2 + y_0^2)(x_1^2 + y_1^2)$), in the instances needed there. Identity (d) is used only in our derivation by hand of Lemma 5; the Lean proof does not need it. □

Lemma 5 (`S_rec`, `Q_rec`)**.** *S and Q satisfy* (1)*: for every $n \geq 0$,*

$$S(n+4) = 2S(n+3) + 2S(n+2) + 2S(n+1) - S(n),$$
$$Q(n+4) = 2Q(n+3) + 2Q(n+2) + 2Q(n+1) - Q(n).$$

Proof. In Lean, $U(n+2), \ldots, U(n+5)$ are unfolded through `step` into integer-linear expressions in the four coordinates x_0, y_0, x_1, y_1 of $U(n)$ and $U(n+1)$; both identities then become polynomial identities in x_0, y_0, x_1, y_1, which `ring` verifies. We give a derivation by hand from Lemma 4.

Put $D(n) = P(n) - Q(n)$. By (a) and (b), $D(n+1) = P(n) + Q(n)$, and hence $D(n+2) = P(n+1) + Q(n+1) = 2S(n+1) + D(n)$. By (d), $2D(n) = T(n)$ with $T(n) = S(n+2) - 2S(n+1) - S(n)$. Therefore $T(n+2) = 2D(n+2) = 4S(n+1) + T(n)$, that is,

$$S(n+4) - 2S(n+3) - S(n+2) = 4S(n+1) + S(n+2) - 2S(n+1) - S(n),$$

which is the recurrence for S. For Q, let $R(n) = Q(n+4) - 2Q(n+3) - 2Q(n+2) - 2Q(n+1) + Q(n)$. By (b), $Q(m) + Q(m+1) = S(m+1)$ for all $m \geq 0$, so $R(n) + R(n+1)$ is the recurrence expression for S at $n+1$, which vanishes. Hence $R(n) = (-1)^n R(0)$, and $R(0) = 12 - 2\cdot 4 - 2\cdot 1 - 2\cdot 1 + 0 = 0$. □

Lemma 6 (`S_eq_Q_eq`)**.** *For every $n \geq 0$, $S(n) = \mathrm{A138573}(n)$ and $Q(n) = \mathrm{A317974}(n+1)$.*

Proof. The Lean proof shows by induction on n that the four equalities $S(n+j) = \mathrm{A138573}(n+j)$, $j = 0, 1, 2, 3$, and the four equalities $Q(n+j) = \mathrm{A317974}(n+j+1)$, $j = 0, 1, 2, 3$, hold. For $n = 0$ these say that $S(0), \ldots, S(3)$ are $0, 1, 2, 5$ and that $Q(0), \ldots, Q(3)$ are $0, 1, 1, 4$, i.e. $\mathrm{A317974}(1), \ldots, \mathrm{A317974}(4)$; the kernel evaluates both sides (`decide`). For the step, the window shifts by one and the new value is supplied by Lemma 5 and the recurrence (1), which A138573 and $n \mapsto \mathrm{A317974}(n+1)$ both satisfy. □

Lemma 7 (`A138573_mono`, `A138573_pos`, `S_pos`)**.** A138573 *is nonnegative and nondecreasing, and* $\mathrm{A138573}(n) \geq 1$ *for* $n \geq 1$*. Hence* $S(n) \neq 0$ *for* $n \geq 1$*.*

Proof. Write $u = \mathrm{A138573}$. We show $0 \leq u(n) \leq u(n+1) \leq u(n+2) \leq u(n+3)$ by induction on n; for $n = 0$ this is $0 \leq 0 \leq 1 \leq 2 \leq 5$. For the step, $0 \leq u(n+1)$ follows from $0 \leq u(n) \leq u(n+1)$, and

$$u(n+4) - u(n+3) = \big(u(n+3) - u(n)\big) + 2u(n+2) + 2u(n+1) \geq 0.$$

Since $u(1) = 1$, monotonicity gives $u(n) \geq 1$ for $n \geq 1$, and $S(n) = u(n)$ by Lemma 6. □

3.2. **The convergents.**

Lemma 8 (`alg`)**.** *Let* $p, q, s, s' \in \mathbb{Q}$ *with* $s, s' \neq 0$ *and* $p^2 + q^2 = ss'$*. Then*

$$(1+i) + \frac{1}{(p+qi)/s} = \frac{(s'+p) + (s'-q)\,i}{s'}.$$

Proof. We have $(p/s)^2 + (q/s)^2 = s'/s \neq 0$, so $1/((p+qi)/s) = (p-qi)/s \cdot s/s' = (p-qi)/s'$, and $(1+i)+(p-qi)/s' = ((s'+p)+(s'-q)i)/s'$. In Lean the statement reads `gadd (1, 1) (ginv ( p / s, q / s)) = ((s' + p) / s', (s' - q) / s')`, and its proof derives $(p/s)^2+(q/s)^2 = s'/s$ by `field_simp` and `linear_combination`, then closes the two coordinates by `field_simp` and `ring`. □

Proposition 9 (`conv_eq`)**.** *For every* $n \geq 1$*,*

$$c_n = \frac{P(n) + Q(n)\,i}{S(n)} = \frac{U(n+1)}{U(n)}.$$

In particular $\operatorname{Im} c_n = Q(n)/S(n) = \mathrm{A317974}(n+1)/\mathrm{A138573}(n)$*.*

Proof. Induction on n. For $n = 1$: $P(1) = Q(1) = S(1) = 1$ (checked by `decide`) and $c_1 = 1 + i$. Suppose the claim holds for $n \geq 1$. Put $p = P(n)$, $q = Q(n)$, $s = S(n)$, $s' = S(n+1)$. Then $s, s' \neq 0$ by Lemma 7 and $p^2 + q^2 = ss'$ by Lemma 4(c), so Lemma 8 gives

$$c_{n+1} = (1+i) + \frac{1}{c_n} = \frac{(S(n+1) + P(n)) + (S(n+1) - Q(n))\,i}{S(n+1)} = \frac{P(n+1) + Q(n+1)\,i}{S(n+1)}$$

by Lemma 4(a),(b). The second form of c_n follows from $W(n)/S(n) = U(n+1)\overline{U(n)}/|U(n)|^2$ (it is not needed in Lean); the last statement follows from Lemma 6. □

3.3. **Proof of Theorem 2.** Let $n \geq 1$ and $c_n = x + yi$, and put $L = \operatorname{den} c_n = \operatorname{lcm}(\operatorname{den} x, \operatorname{den} y)$, which is positive. Since $\operatorname{den} y \mid L$,

$$\frac{a(n)}{\mathrm{A093727}(n)} = \frac{\operatorname{num} y \cdot (L/\operatorname{den} y)}{L} = \frac{\operatorname{num} y}{\operatorname{den} y} = y = \operatorname{Im} c_n.$$

In Lean this is the step `hl` of `result`, using `Nat.dvd_lcm_right`, `Nat.cast_div` and `Rat.num_div_den`. By Proposition 9, $\operatorname{Im} c_n = \mathrm{A317974}(n+1)/\mathrm{A138573}(n)$. □

Remark 10 (Consequences, not formalized)**.** (i) Lemma 4(b) and Lemma 6 give $\mathrm{A317974}(n) + \mathrm{A317974}(n+1) = \mathrm{A138573}(n)$ for $n \geq 0$; this also follows directly from the generating functions $x^2(1-x)/(1-2x-2x^2-2x^3+x^4)$ and $x(1-x)(1+x)/(1-2x-2x^2-2x^3+x^4)$ recorded in the two entries. (ii) By Lemma 4(a) and $P(0) = 0$, $\operatorname{Re} c_n = P(n)/S(n) = \big(\sum_{k=1}^{n} \mathrm{A138573}(k)\big)/\mathrm{A138573}(n)$. (iii) With $g_n = \gcd(P(n), Q(n), S(n))$, one has $\mathrm{A093727}(n) = S(n)/g_n$ (because $dc_n \in \mathbb{Z}[i]$ iff $S(n) \mid dP(n)$ and $S(n) \mid dQ(n)$, iff $S(n)/g_n \mid d$), $a(n) = Q(n)/g_n$ and $\mathrm{A093725}(n) = P(n)/g_n$. For example $g_4 = 4$, $g_5 = 3$, and $g_{42} = 15457$.

4. Machine verification

Files and audit. The proof is the file `lean/Math0/Frontier/OEIS/A093726.lean` of the math0 repository, `https://github.com/Foundation-0/math0`. The main declaration is `Math0.Frontier.A093726.result`, of type `Math0.Statements.A093726.statement`; the auxiliary definitions `step`, `U`, `S`, `P`, `Q` and the lemmas `S_rec`, `Q_rec`, `S_eq_Q_eq`, `A138573_mono`, `A138573_pos`, `S_pos`, `alg`, `conv_eq` are private. The file uses no `sorry`, `native_decide`, `axiom` or `set_option`; the only finite computations are the base cases, closed by `decide`, which the kernel checks. The frozen statement file was not modified. The proof file ends with

```
#math0_settles Math0.Frontier.A093726.result Math0.Statements.A093726.statement
```

a command defined in `lean/Math0/Audit.lean` for statements formalized by math0. It fails unless `statement` is a `Prop`-valued definition without universe parameters, the type of `result` is syntactically `statement` or its negation (up to metadata, without unfolding), and both depend only on the axioms `propext`, `Classical.choice` and `Quot.sound` (which excludes `sorryAx`, `native_decide` and custom axioms). Compiling the file with `lake env lean` reports:

```
math0 audit: 'Math0.Statements.A093726.statement' depends on axioms [Classical.choice, Quot.sound, propext]
math0 audit: 'Math0.Frontier.A093726.result' depends on axioms [Classical.choice, Quot.sound, propext]
math0 statement check: 'Math0.Frontier.A093726.result' proves 'Math0.Statements.A093726.statement'
```

Toolchain and reproduction. We use Lean 4 [5] `v4.33.1` and Mathlib [4] `v4.33.1`, pinned in `lean/`; the repository also pins formal-conjectures [2] at commit `c77fdd4`, which this result does not use (the statement is math0's own formalization). To reproduce, run `scripts/verify.sh`: it fetches the Mathlib cache, builds the library with `lake build` (re-running every audit), checks that the pinned `formal_conjectures` and `mathlib` checkouts are unmodified, and counts the passed statement checks.

References

[1] H. S. M. Coxeter, *Loxodromic sequences of tangent spheres*, Aequationes Math. **1** (1968), 104–121.

[2] M. Firsching, P. Lezeau, S. Mercuri, M. Z. Horváth, Y. Dillies, C. Sönne, E. Wieser, F. Zhang, T. Hubert, B. Agüera y Arcas, and P. Kohli, *Formal Conjectures: An Open and Evolving Benchmark for Verified Discovery in Mathematics*, arXiv:2605.13171, 2026. Repository: `https://github.com/google-deepmind/formal-conjectures`.

[3] S. Fried, *Further proofs of conjectures from the OEIS*, arXiv:2607.24832, 2026.

[4] The mathlib Community, *The Lean mathematical library*, in: Proceedings of the 9th ACM SIGPLAN International Conference on Certified Programs and Proofs (CPP 2020), ACM, 2020, 367–381.

[5] L. de Moura and S. Ullrich, *The Lean 4 theorem prover and programming language*, in: Automated Deduction – CADE 28, Lecture Notes in Comput. Sci. **12699**, Springer, 2021, 625–635.

[6] The OEIS Foundation Inc., A093725 and A093727 (R. G. Wilson v, Mar. 11, 2004), `https://oeis.org/A093725`, `https://oeis.org/A093727`.

[7] The OEIS Foundation Inc., A093726 (R. G. Wilson v, Mar. 11, 2004; definition corrected and conjecture added by Z. He, Jun. 16, 2026), `https://oeis.org/A093726`.

[8] The OEIS Foundation Inc., A138573 (B. Cloitre, May 12, 2008; formulas by P. Bala, Mar. 4, 2014), `https://oeis.org/A138573`.

[9] The OEIS Foundation Inc., A317974 (N. J. A. Sloane, Sep. 3, 2018), `https://oeis.org/A317974`.

Produced by Claude (an AI model developed by Anthropic) in the math0 project